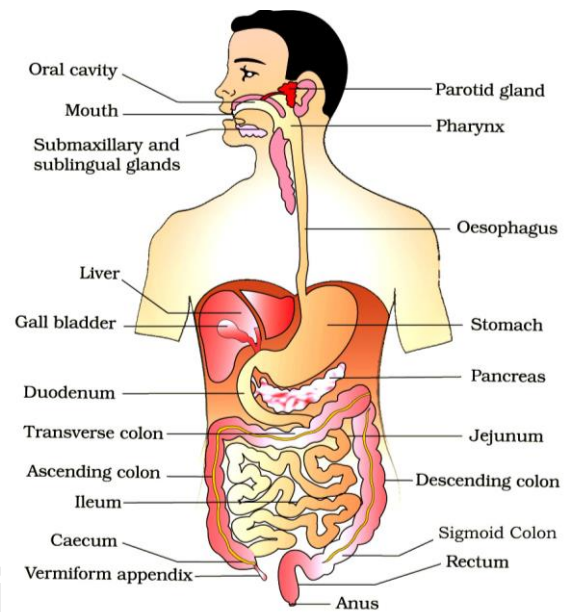
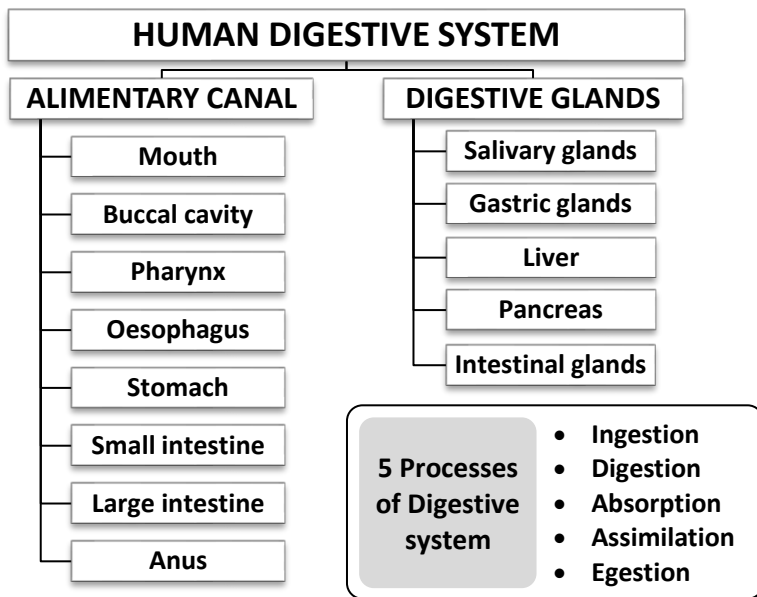


16. DIGESTION AND ABSORPTION

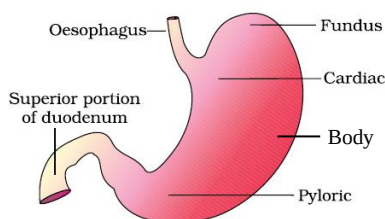
- **Nutrition** is the getting and utilization of energy rich nutrients (food) by an organism.
- Food consists of **carbohydrates, proteins, fats (lipids), vitamins, minerals** and **water**.
- Food provides **energy for life activities, materials for growth, maintains body temperature** and **repairs tissues**.
- The water plays an important role in metabolic processes and prevents dehydration of the body.



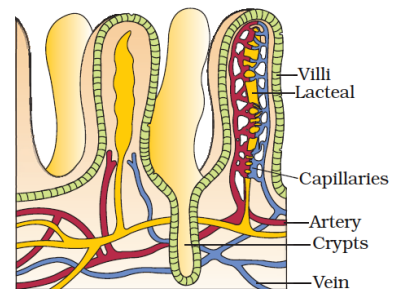
I. ALIMENTARY CANAL (GUT)

It includes the following parts:

- **Mouth:** To receive the food (**ingestion**).
- **Buccal cavity (oral or mouth cavity):**
 - Consists of **palate (roof), teeth & muscular tongue**.
 - Palate has anterior **hard palate** and posterior **soft palate**.
 - Tongue is a muscular organ attached to the floor of oral cavity by the **frenulum**. Tongue has small projections called **papillae**. Some papillae bear **taste buds**.
 - At the back, on either side of tongue **tonsils** present.
- **Pharynx:**
 - Common passage for digestive and respiratory systems.
 - When food materials pass through the pharynx, the cartilaginous **epiglottis** closes the **glottis** (opening of larynx) and prevents the entry of food into trachea.
- **Oesophagus:**
 - Muscular tube (30 cm) that conducts food into stomach.
 - Posterior part of the oesophagus has **gastro-oesophageal sphincter** (a circular muscle). It controls the opening of oesophagus into stomach.
- **Stomach:**
 - 'J' shaped structure for storage and digestion of food.
 - **4 parts:** a **cardiac** portion into which the oesophagus opens, a **fundic** region, **body** (main central region) and a **pyloric** portion (antrum).
 - Pyloric stomach leads to small intestine by an opening called **Pylorus**, guarded by **pyloric sphincter muscle**.
 - Inner wall of stomach bears **rugae** (longitudinal folds).



- **Small intestine:**
 - Longest part of gut (7 m long and 2.5 cm diameter).
 - Consists 3 parts, namely **duodenum** (C shaped first part), **Jejunum** (middle part) and **Ileum** (terminal part).
 - Finger-like **villi** are seen at the mucosa. Each villus has a brush-bordered columnar epithelial layer provided with **microvilli**. Villus consists of a capillary network and a small lymph vessel (**lacteal**).
- **Large intestine:**
 - 1.5 m long. Consists of **caecum, colon** and **rectum**.
 - Caecum is well-developed in herbivores but very small in man. Arising from the caecum is a finger-like vestigial organ, the **vermiform appendix**.
 - The colon consists of **ascending colon, transverse colon, descending colon** and **sigmoid colon**.
 - Pelvic colon leads to **rectum** that opens out by **anus**. Anus is guarded by **anal sphincter** (circular voluntary muscles).
 - In some herbivores, the large intestine consists of several cellulose digesting bacteria.



TEETH

Nature & mode of arrangement of teeth is called **dentition**. **Human dentition** is Thecodont, Heterodont & Diphyodont.

- **Thecodont:** It means teeth are placed in the jaw sockets.
- **Heterodont:** It means different kinds of teeth are present. They are **incisors (I)** for cutting, **canines (C)** for tearing, **premolars (PM)** & **molars (M)** for mastication.

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